

```
public abstract class Employee
{
    private String empID;
    public String name;
    private String pan;
    public String joiningDate;
    public String designation;

    public String getEmpID()
    {
        return empID;
    }
    public void setEmpID(String empID)
    {
        this.empID = empID;
    }

    public String getPAN()
    {
        return pan;
    }
    public void setPAN(String pan)
    {
        this.pan = pan;
    }

    public abstract float calcBenefits(Payroll p);
    public abstract float calcCTC(Payroll p);
}

public class Payroll
{
    public int noOfHours;
    public float hourlyRates;
    public float baseSalary;
    public float perfBonus;

    public int insurance;
    public int healthInsurance;
    public float pf;
    public int pto;
}
```

```

public class FullTimeEmployee extends Employee
{
    public float calcBenefits(Payroll p)
    {
        // full-time benefits
        float benefits = p.insurance + p.healthInsurance + p.pf;
        return benefits;
    }

    public float calcCTC(Payroll p)
    {
        return p.baseSalary + p.perfBonus + calcBenefits(p);
    }
}

public class Manager extends FullTimeEmployee
{
    public float travelAllowance = 10000;
    public float educationAllowance = 5000;

    public float calcBenefits(Payroll p)
    {
        float extra = 10000; // manager-specific benefit
        return super.calcBenefits(p) + extra;
    }

    public float calcCTC(Payroll p)
    {
        return p.baseSalary + p.perfBonus + travelAllowance +
educationAllowance + calcBenefits(p);
    }
}

public class ContractEmployee extends Employee
{
    public float calcBenefits(Payroll p)
    {
        // minimal benefits
        return 2000;
    }

    public float calcCTC(Payroll p)
    {
        return (p.noOfHours * p.hourlyRates) + calcBenefits(p);
    }
}

```

```

public class MainForEmployee
{
    public static void main(String[] args)
    {
        Payroll p1 = new Payroll();
        p1.baseSalary = 50000;
        p1.perfBonus = 10000;
        p1.insurance = 5000;
        p1.healthInsurance = 3000;
        p1.pf = 6000;

        FullTimeEmployee e = new FullTimeEmployee();
        System.out.println("Full Time Employee CTC:" + e.calcCTC(p1));

        Payroll p2 = new Payroll();
        p2.baseSalary = 80000;
        p2.perfBonus = 20000;
        p2.insurance = 7000;
        p2.healthInsurance = 4000;
        p2.pf = 8000;

        Manager m = new Manager();
        System.out.println("Manager CTC:" + m.calcCTC(p2));

        Payroll p3 = new Payroll();
        p3.noOfHours = 160;
        p3.hourlyRates = 500;

        ContractEmployee ce = new ContractEmployee();
        System.out.println("Contract Employee CTC:" + ce.calcCTC(p3));
    }
}

```

PROBLEMS (2) OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code + - [] [] ... | X

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PS C:\Users\hp\Downloads\Programming in Java> cd "c:\Users\hp\Downloads\Programming in Java\" ; if ($?) { javac MainForEmployee.java } ; if ($?) { java MainForEmployee }
Full Time Employee CTC:74000.0
Manager CTC:144000.0
Contract Employee CTC:82000.0
PS C:\Users\hp\Downloads\Programming in Java>

```